

Econ 144: Project 1

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Packages

```
library(readr) ## reading in data
library(ggplot2) ## graphing
library(patchwork) ## displays multiple ggplots in one window
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method              from
##   as.zoo.data.frame zoo
```

Data

```
data <- read_csv("australia_housing_prices.csv")

## Rows: 222 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbl  (1): QAUR628BIS
## date (1): observation_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Part 1: Introduction

```
head(data)

## # A tibble: 6 x 2
##   observation_date QAUR628BIS
##   <date>          <dbl>
## 1 1970-01-01      31.3
## 2 1970-04-01      31.6
## 3 1970-07-01      32.1
## 4 1970-10-01      32.4
## 5 1971-01-01      33.0
## 6 1971-04-01      33.4
```

For this project, we will be using a FRED dataset that includes quarterly data on the Real Residential Home Prices in Australia. The data is normalized to Q1 of 2010, which equals 100, and the data is not yet seasonally adjusted. The first observation was taken in Q1 of 1970, and the most recent was released on September 25 of this year (Q3). There are 222 observations in total, and each obs has a date and a real home price figure.

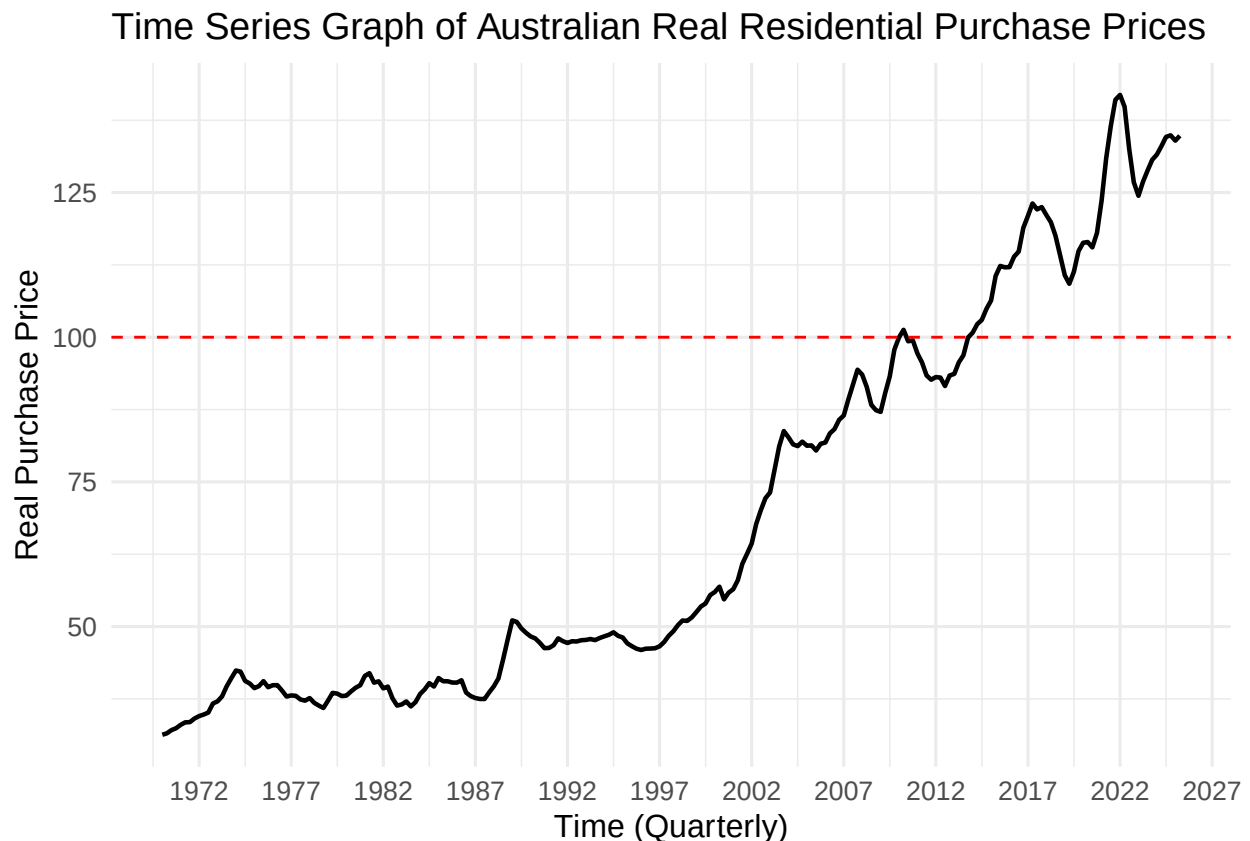
Being a first world country in a thriving economy, we expect home prices to steadily be increasing over time, but don't have a great idea of how seasonality will impact sales. We know that home buyers have purchasing preferences in America (less popular to buy a home in winter, for example), but due to their opposite seasons, this trend may not hold in Australia.

Part 2: Results

1

a)

```
ggplot(data, aes(x = observation_date, y = QAUR628BIS)) +  
  geom_line(linewidth = 0.8) + ## aesthetic preference  
  geom_hline(yintercept = 100, linetype = "dashed", color = "red") + ##not necessary but added to show  
  labs( ## titles  
    title = "Time Series Graph of Australian Real Residential Purchase Prices",  
    x = "Time (Quarterly)",  
    y = "Real Purchase Price"  
  ) +  
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") + ## aesthetic preference  
  theme_minimal(base_size = 12) ## aesthetic preference
```



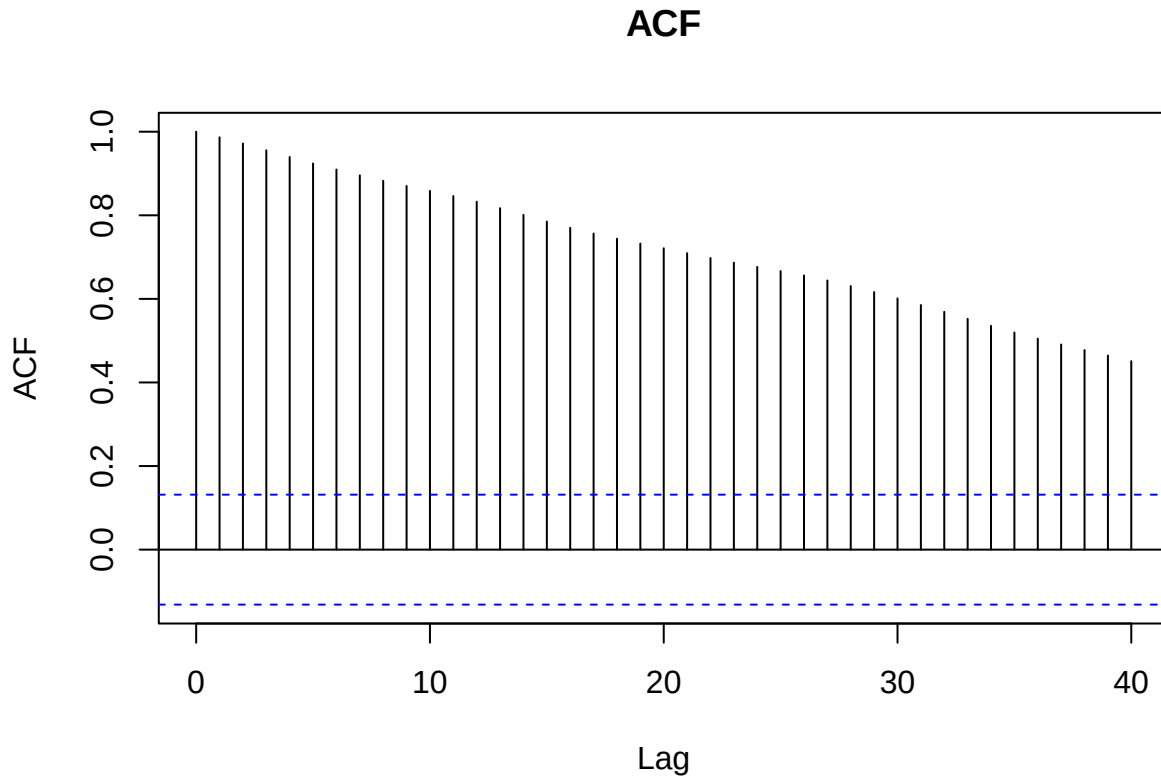
The above graph is a simple time series plot, with a red dashed line at $y = 100$ to show the level to which the rest of the data is normalized (Q1 2010).

b)

This data is not covariance stationary. For a graph to be covariance stationary, it needs to exhibit constant mean, variance and autocovariance (covariance only caused by lag) over time. Due to the upward trending nature of this graph, the mean isn't constant, and therefore the graph isn't covariance stationary.

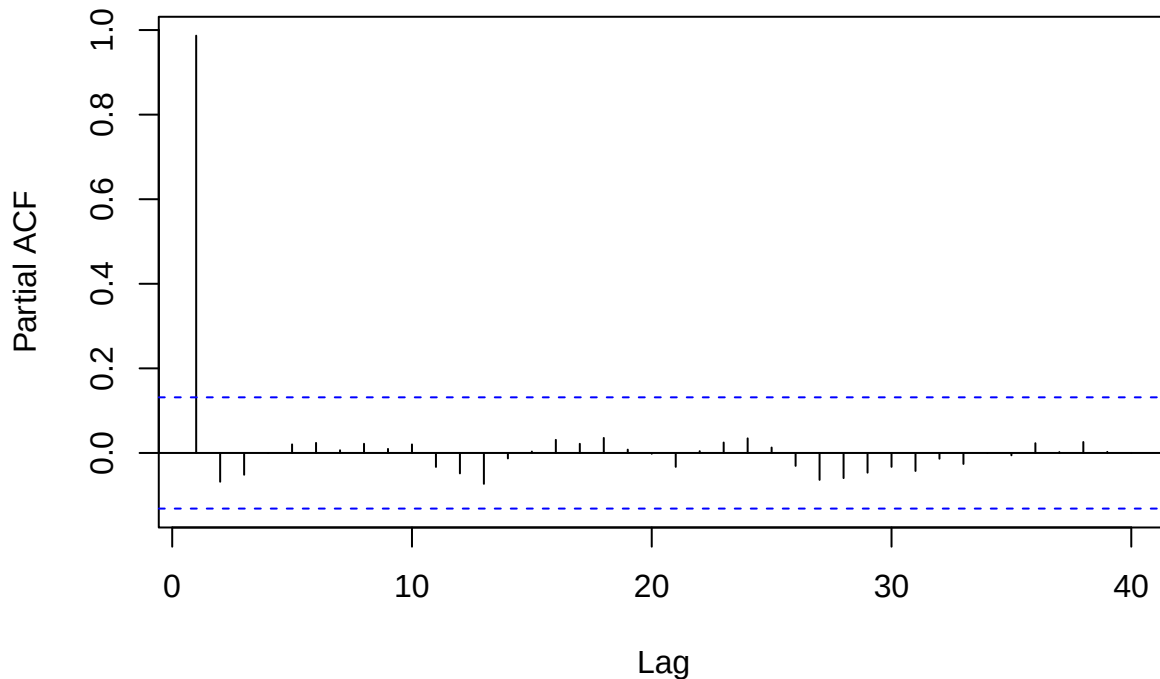
c)

```
y <- data$QAUR628BIS ## renaming for convenience
acf(y, lag.max = 40, main = "ACF") ## base r acf plot
```



```
pacf(y, lag.max = 40, main = "PACF") ## base r pacf plot
```

PACF



ACF:

The ACF plot shows strong correlations up to (and presumably past) lag 40. A typical ACF plot would exhibit some correlation in the first few lags but then would typically fall below the dotted blue line. Since this does not occur in the data, it may prove useful to take the log of our data (which tends to make it more stationary) and recalculate the ACF plot.

PACF: Once the linearity affects were removed from the ACF plot, the steady decline in correlation went away and was replaced with one very strong spike after lag 1. The rest of the lags never cross the blue significance line, but they do move in a sin wave esque pattern, which may suggest a small seasonal or year-to-year correlation. Removing seasonality may make the PACF and the ACF plots behave more in line with expected plots.

d)

```
t <- seq_along(y) ## added for the linear trend, so that we can relate real price to quarter, not date

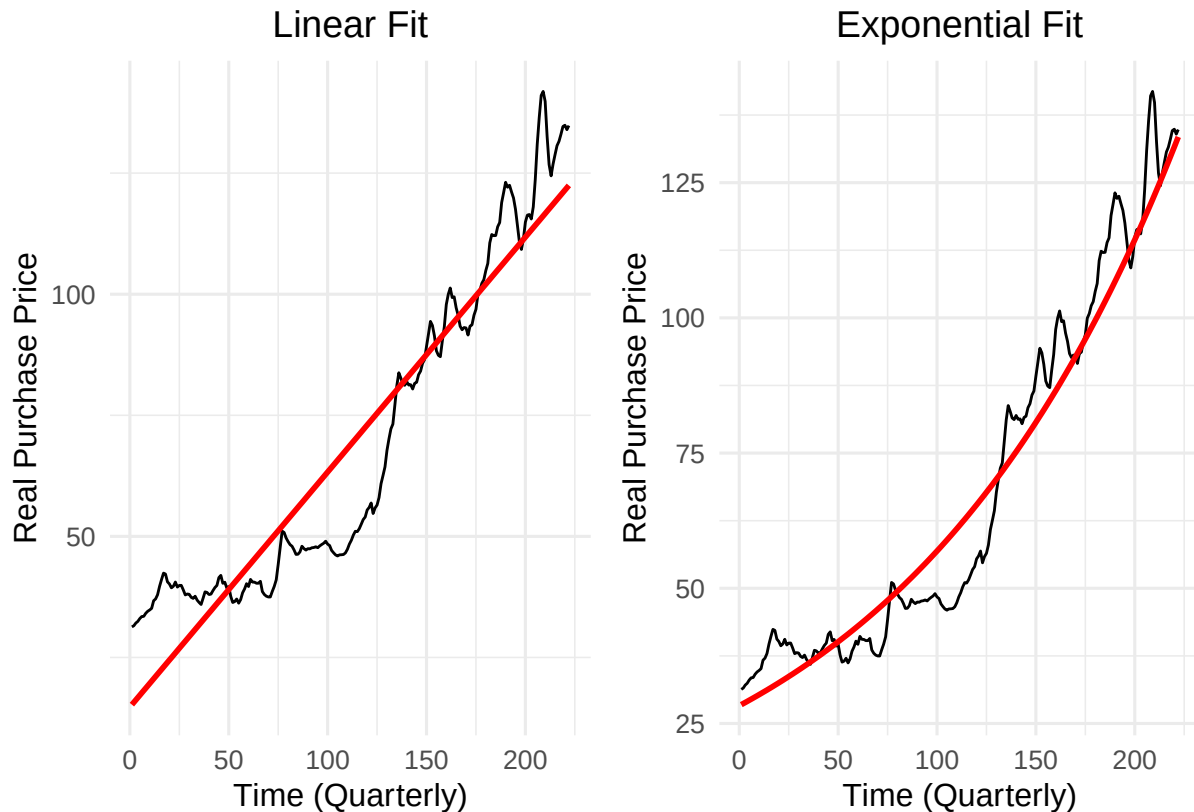
m1 <- lm(y ~ t) ## linear regression
m2 <- lm(log(y) ~ t) ## exponential (nonlinear) regression

df <- data.frame(t = t, y = y) ## puts the y and t variables into one dataframe for graphing purposes
df$fit_lin <- fitted(m1) ## generates predicted data from models
df$fit_exp <- exp(fitted(m2))

p_lin <- ggplot(df, aes(t, y)) +
  geom_line() +
  geom_line(aes(y = fit_lin), linewidth = 1, color = "red") +
  labs(title = "Linear Fit",
       x = "Time (Quarterly)", y = "Real Purchase Price") + ## aesthetic preference
  theme_minimal(base_size = 12) + ## aesthetic preference
  theme(plot.title = element_text(hjust = 0.5)) ## centered title
```

```
p_exp <- ggplot(df, aes(t, y)) +
  geom_line() +
  geom_line(aes(y = fit_exp), linewidth = 1, color = "red") +
  labs(title = "Exponential Fit",
       x = "Time (Quarterly)", y = "Real Purchase Price") + ## aesthetic preference
  theme_minimal(base_size = 12) + ## aesthetic preference
  theme(plot.title = element_text(hjust = 0.5)) ## centered title

(p_lin | p_exp) ## patchwork way to display multiple ggplots
```



e)

```
dfm1 <- data.frame(fit = fitted(m1), res = resid(m1))
dfm2 <- data.frame(fit = fitted(m2), res = resid(m2))

res_lin <- ggplot(dfm1, aes(x = fit, y = res)) +
  geom_point(alpha = 0.6) + ## opacity
  geom_hline(yintercept = 0, color = "red") + ## middle line
  labs(x = "Fitted values (level)", y = "Residuals (level)",
       title = "Residuals vs Fitted: Linear") + ## aesthetic preference
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(hjust = 0.5)) ## centered title

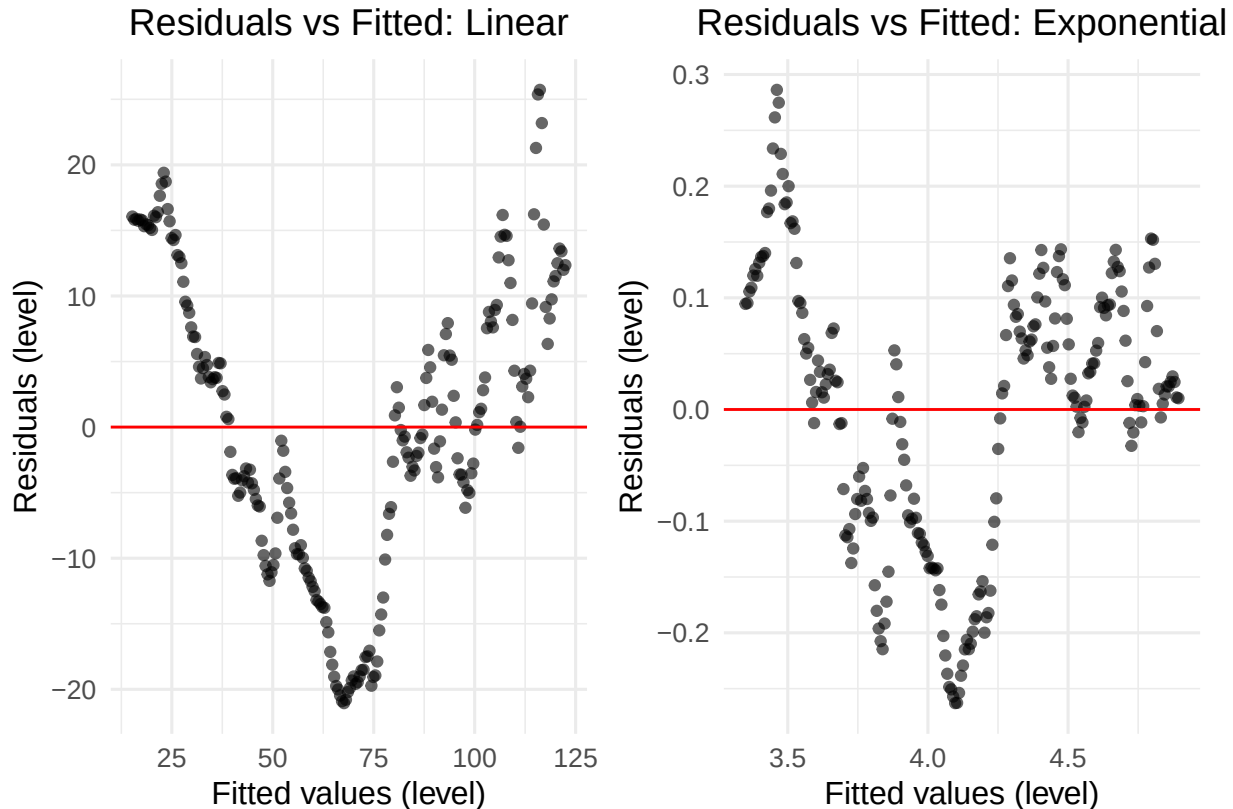
res_exp <- ggplot(dfm2, aes(x = fit, y = res)) +
  geom_point(alpha = 0.6) + ## opacity
  geom_hline(yintercept = 0, color = "red") + ## middle line
  labs(x = "Fitted values (level)", y = "Residuals (level)",
```

```

title = "Residuals vs Fitted: Exponential") + ## aesthetic preference
theme_minimal(base_size = 12) +
theme(plot.title = element_text(hjust = 0.5)) ## centered title

(res_lin | res_exp) # patchwork

```



The above residual vs fitted values plots show that there are underlying issues in our models. You would hope to see scattered randomness in these plots, but instead see clear patterns. The linear model's residuals are clearly “V” shaped, implying that a simple linear regression will significantly over or under predict based on the time of the data. The exponential model (right hand side) isn't much better. The first half of the graph looks much like the linear model. The second half consistently under predicts the data. Although the visual fit was promising, these graphs shows that our models have underlying issues that need to be addressed.

f)

```

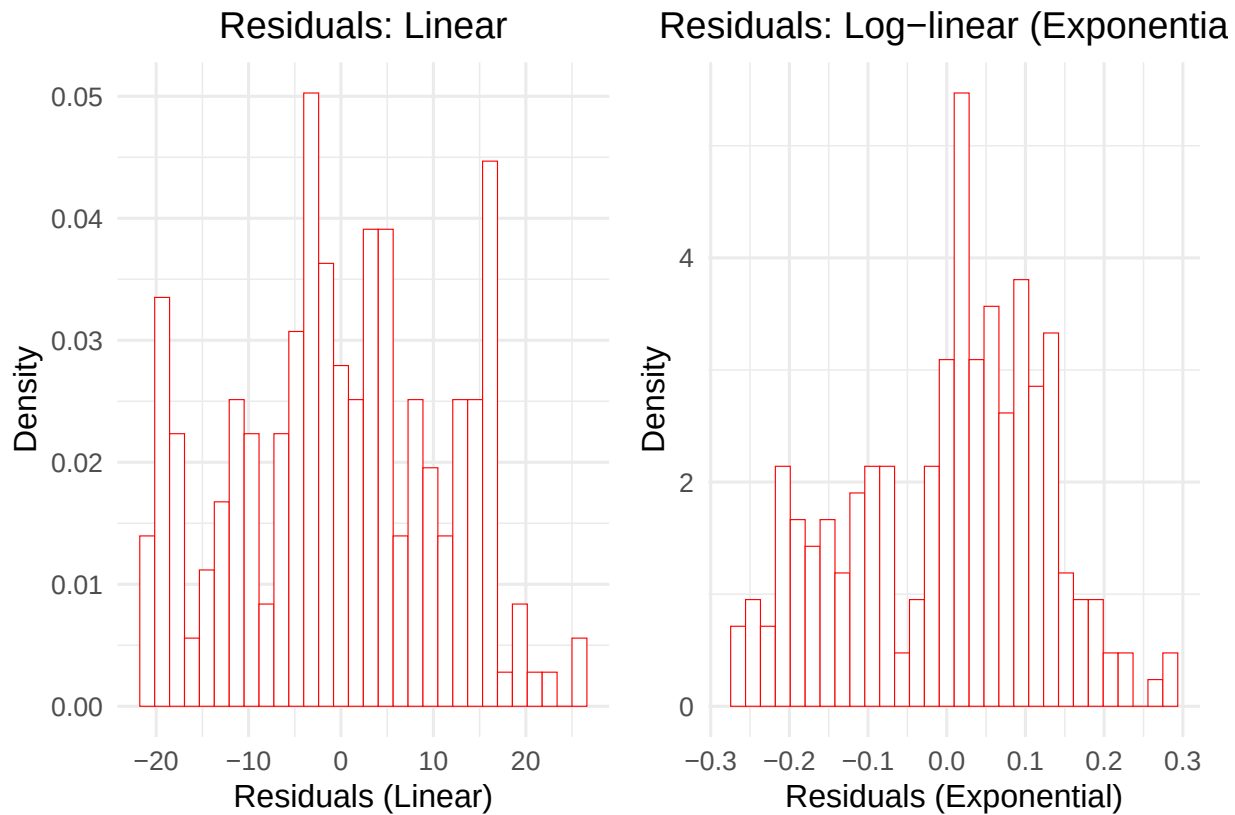
hist_lin <- ggplot(dfm1, aes(x = res)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 30, fill = "white", color = "red", linewidth = 0.2) + ## aesthetic
  labs(title = "Residuals: Linear",
    x = "Residuals (Linear)", y = "Density") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(hjust = 0.5)) ## centered title

hist_exp <- ggplot(dfm2, aes(x = res)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 30, fill = "white", color = "red", linewidth = 0.2) + ## aesthetic
  labs(title = "Residuals: Log-linear (Exponential)",
    x = "Residuals (Exponential)", y = "Density") +

```

```
theme_minimal(base_size = 12) +
theme(plot.title = element_text(hjust = 0.5)) ## centered title

(hist_lin | hist_exp)
```



These histograms show a different view of the residuals than the residuals vs fitted plots. They tend to be centered around 0, and the Exponential model residuals have a bell curve esque feel to them, which is exactly what we want to see. The Linear model residuals are less promising, with large spikes on the tails. These graphs confirm that the linear model is not a great fit and strengthens the argument in favor of the exponential graph despite a less-than-perfect set of residuals.

g)

```
summary(m1)

##
## Call:
## lm(formula = y ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.0543  -8.1229  -0.0836   8.6088  25.7197
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  14.76344    1.52349   9.691  <2e-16 ***
## t             0.48513    0.01185  40.952  <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.31 on 220 degrees of freedom
## Multiple R-squared:  0.884, Adjusted R-squared:  0.8835
## F-statistic: 1677 on 1 and 220 DF, p-value: < 2.2e-16
```

```
summary(m2)
```

```
##
## Call:
## lm(formula = log(y) ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.26297 -0.09760  0.02093  0.09321  0.28624
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.3422834  0.0166563  200.66  <2e-16 ***
## t           0.0069889  0.0001295   53.96  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1237 on 220 degrees of freedom
## Multiple R-squared:  0.9298, Adjusted R-squared:  0.9294
## F-statistic: 2912 on 1 and 220 DF, p-value: < 2.2e-16
```

R-squared: The R-squared statistic, which generally shows how variance is explained by a model, is valued between 0 and 1, with 1 being a perfect, error-free fit and 0 meaning no correlation at all. The Linear model gives an R-squared value of 0.8835, which is incredibly high, but less than the exponential model, which has an R-squared of 0.9294. By this metric, the exponential model should better explain variance than the linear model.

T-statistic: The t-stat, which shows the probability that of a non-zero trend, is high in both models (Linear: 40.952, Exponential: 53.96). In both cases, the quarter (t) is very significant to the Real Home Price in Australia, but odds of being able to reject the slope = 0 null hypothesis are higher in the second model.

F-Statistic: This stat measures trends, where the higher values allow you to reject a no-trend null hypothesis. In our case, both models allow you to reject the null hypothesis and claim that there exists some relationship between our predictor and Real Home Price in Australia. The F-stat is higher in the exponential model (2912 vs 1677 on the same degrees of freedom), but at these levels the actual number does not make a huge statistical difference.

h)

```
AIC(m1)
```

```
## [1] 1711.061
```

```
AIC(m2)
```

```
## [1] -294.0296
```

```
BIC(m1)
```

```
## [1] 1721.269
```



```
BIC(m2)
```

```
## [1] -283.8216
```

When it comes to AIC and BIC, lower is typically better. In our case, the AIC and BIC figures both consistently show that m2 is the better model. This backs up our graphical and statistical findings from the previous sections.

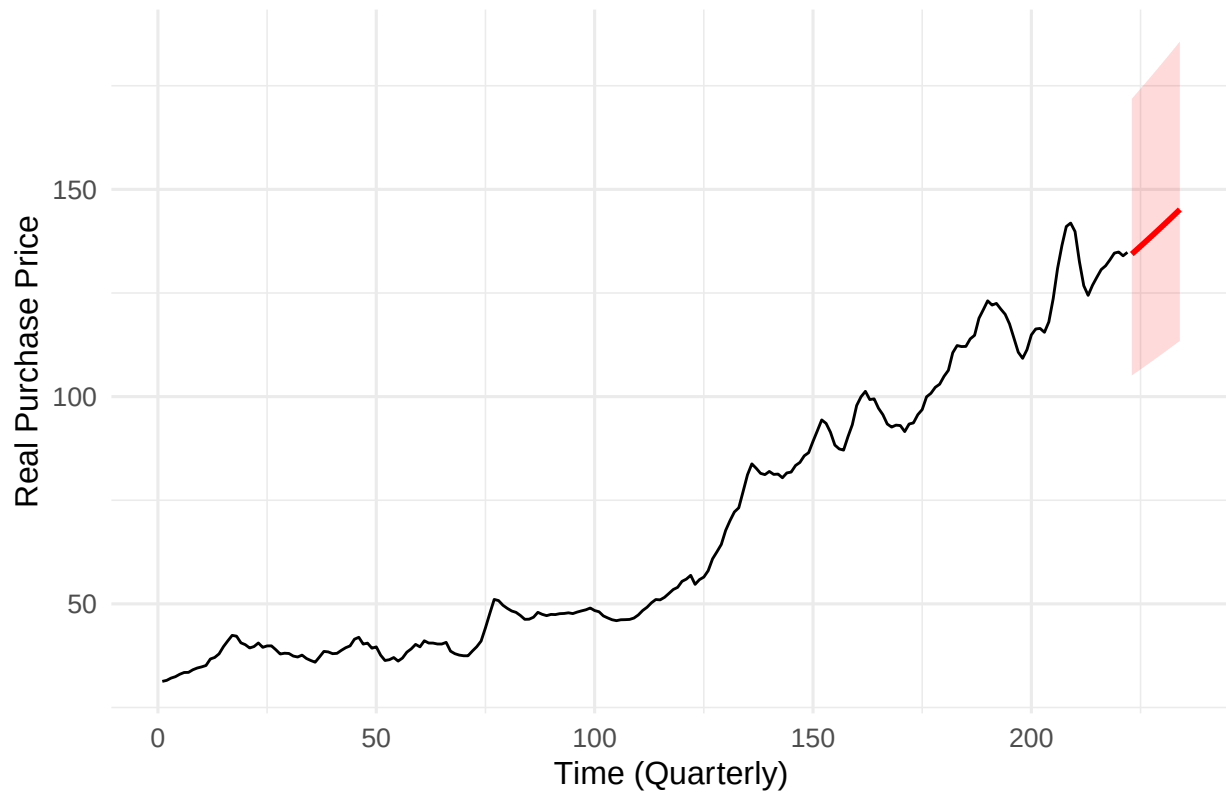
i)

```
h <- 12 ## future periods (measured in quarters)
t_future <- max(t) + seq_len(h) ##adding in data

pred <- predict(m2, newdata = data.frame(t = t_future),
               interval = "prediction", level = 0.95) ## predicting new points
fc <- data.frame( ## new data
  t      = t_future,
  mean   = exp(pred[, "fit"]), ## expected value
  lower  = exp(pred[, "lwr"]), ##lower bound
  upper  = exp(pred[, "upr"]) ## upper bound
)

ggplot() +
  geom_line(data = df, aes(t, y)) + ## plotting the historical data
  geom_ribbon(data = fc, aes(t, ymin = lower, ymax = upper), fill = "red", alpha = 0.15) + ## adding th
  geom_line(data = fc, aes(t, mean), color = "red", linewidth = 1) + ## predicted data
  labs(title = "Log-Linear 12-Quarter Forecast",
       x = "Time (Quarterly)", y = "Real Purchase Price") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(hjust = 0.5)) ## centered title
```

Log-Linear 12-Quarter Forecast



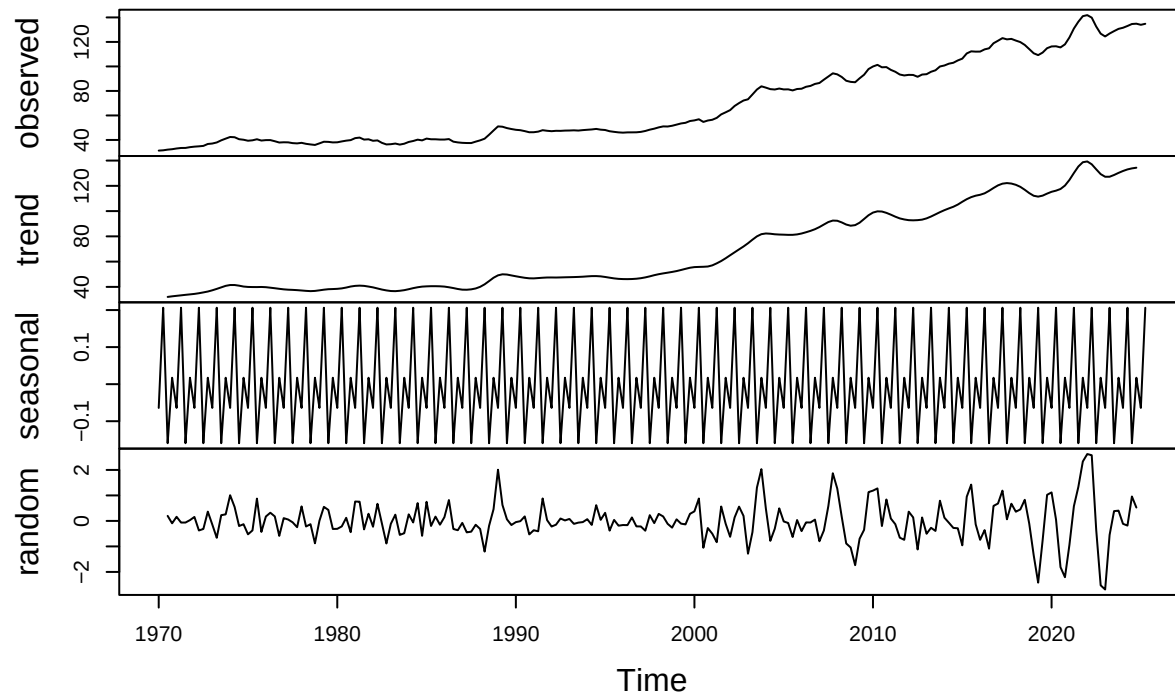
The above plot shows the quarterly time series data extrapolated 12 extra quarters based on $m2$ ($\ln(\log(y) \sim t)$). The red shaded portion is the certainty bounds, with the middle red bar showing the predicted values of the next 12 quarters.

2.

a)

```
y <- ts(y, start=c(1970,1), frequency = 4) ## make data into time series object
add_dec <- decompose(y, type = "additive")
plot(add_dec) ## plot add decomp
```

Decomposition of additive time series



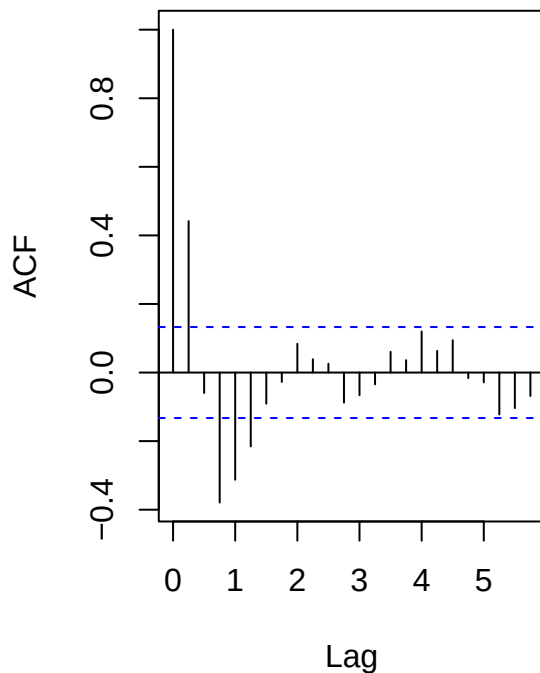
```
resid_add <- y - add_dec$seasonal - add_dec$trend ## get residuals after removing seasonality and trend
```

```
par(mfrow=c(1,2))
```

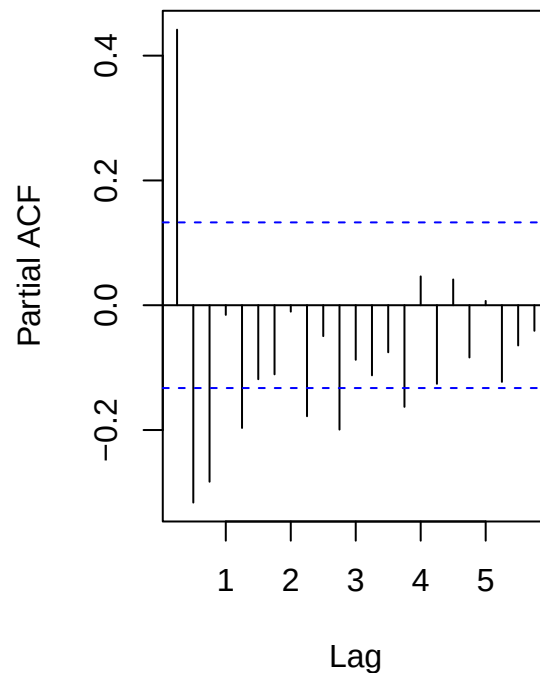
```
acf(na.omit(resid_add), main="ACF: additive residuals") ## plot acf of residuals
```

```
pacf(na.omit(resid_add), main="PACF: additive residuals") ## plot pacf of residuals
```

ACF: additive residuals



PACF: additive residuals

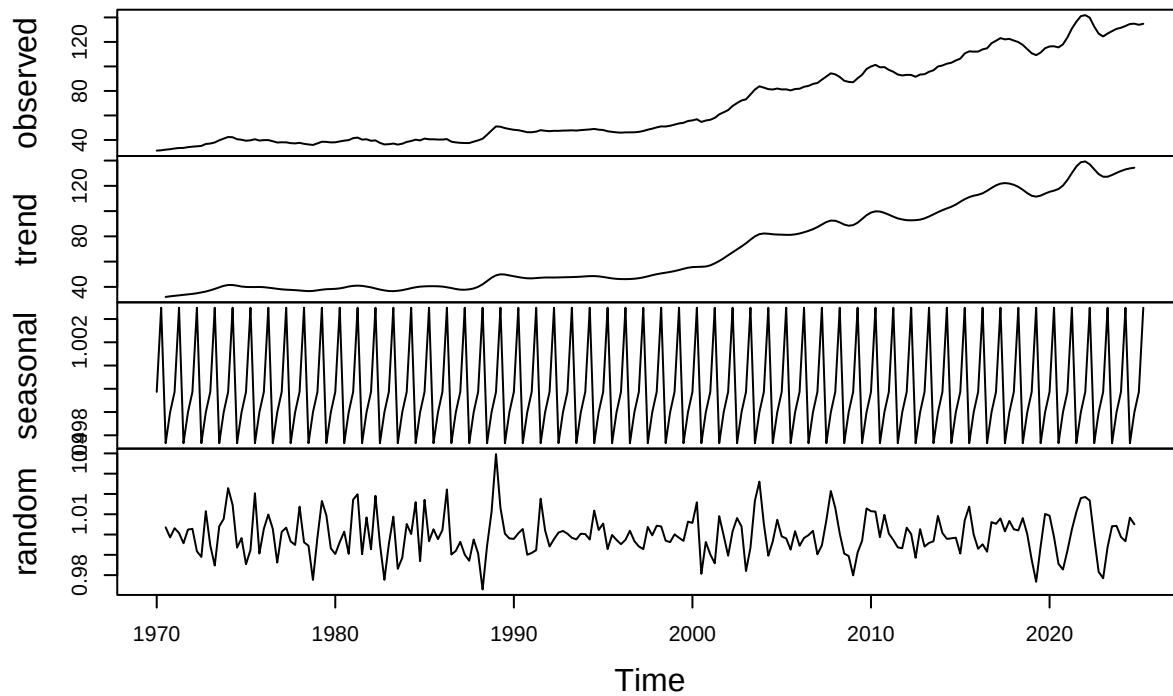


In both the ACF and the PACF we see a massive spike at lag 1, then smaller but noticeable correlations at subsequent lags. This suggests that there is some serial correlation in the data even after the removal of trend and seasonality (the residuals aren't just white noise) and that there is still some short-term dependence structure.

b)

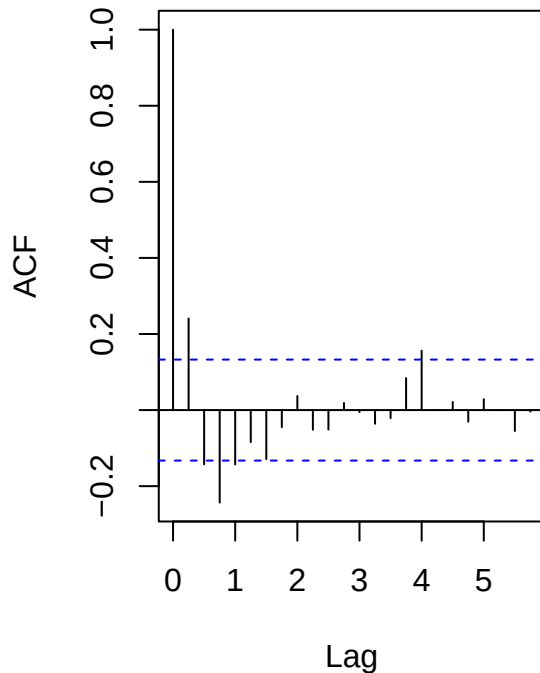
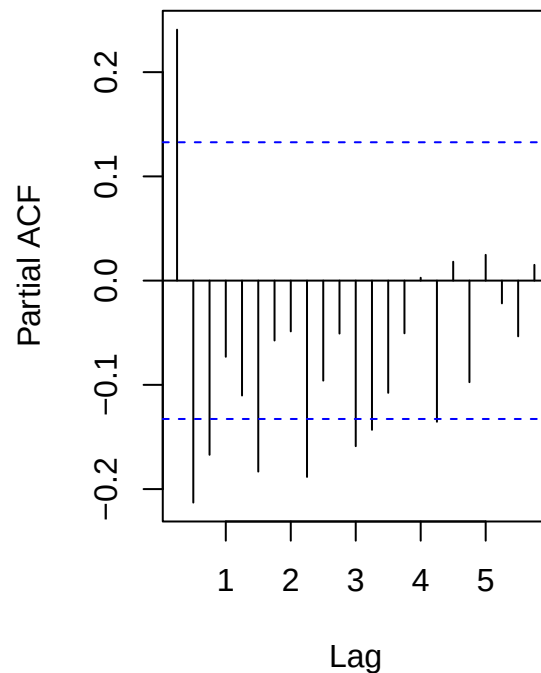
```
mult_dec <- decompose(y, type = "multiplicative")  
plot(mult_dec) ## plot multiplicative decom
```

Decomposition of multiplicative time series



```
resid_mult <- na.omit(mult_dec$random) ##  $y = \text{trend} \times \text{seasonal} \times \text{random}$ ,  $\text{random} = y / (\text{seasonal} \times \text{trend})$ 

par(mfrow=c(1,2))
acf(na.omit(resid_mult), main="ACF: multiplicative residuals")
pacf(na.omit(resid_mult), main="PACF: multiplicative residuals")
```

ACF: multiplicative residuals**PACF: multiplicative residuals**

In the ACF plot, the first lag shows strong autocorrelation, and soon after the correlations drop to near 0, meaning there's still a bit of short-term dependence, but the correlations past the first few lags are weak, like the additive decomposition. The PACF is also similar to the additive model, with strong correlations in the first three lags, then some potential outlier correlations at later lags.

c

In our opinion, since the additive and multiplicative models are very similar in shape and correlation magnitude, and the multiplicative decomposition is not adding anything significant materially, it makes more sense to use the simpler additive decomposition, therefore, in this case, additive is “better”.

d

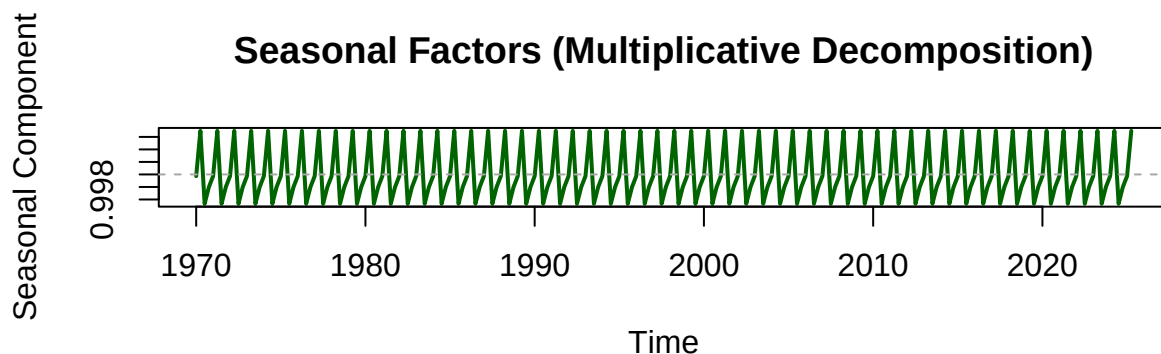
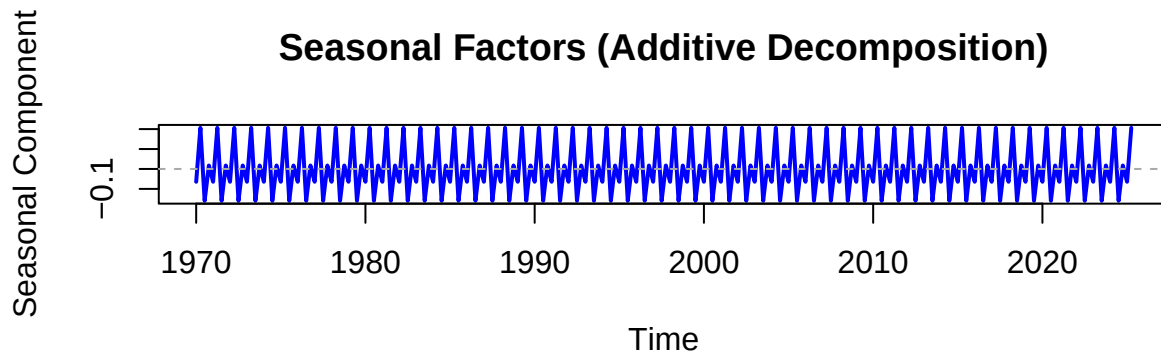
Based on these two decomposition and interpretation of random components, our models for the cycles would be very similar because in both the additive and multiplicative cases, the random (cycle) component shows a short-term dependence (spike at lag 1, i.e. AR(1) behavior) implying that the cyclical behavior of housing prices has a short-memory-like process. Therefore, since the residuals are nearly identical in the two decomposition, the underlying cyclical model would also be structurally very similar.

e

```
add_dec <- decompose(y, type="additive") ## same decomp
mult_dec <- decompose(y, type="multiplicative") ## same decomp

par(mfrow = c(2,1))
plot(add_dec$seasonal, ## plot for additive seasonal factors
     main = "Seasonal Factors (Additive Decomposition)",
     ylab = "Seasonal Component", xlab = "Time", col = "blue", lwd = 2)
abline(h = 0, col = "darkgray", lty = 2)
```

```
plot(mult_dec$seasonal, ## plot for multiplicative seasonal factors
     main = "Seasonal Factors (Multiplicative Decomposition)",
     ylab = "Seasonal Component", xlab = "Time", col = "darkgreen", lwd = 2)
abline(h = 1, col = "darkgray", lty = 2)
```



The additive seasonal factors plot oscillate around 0 (-0.1), which shows the absolute seasonal effect on housing prices each year is pretty weak/negligible. It follows generally the same pattern of peaks and troughs, indicating that seasonal influence on housing prices has not changed or intensified over time. The multiplicative plot oscillates around 1, meaning housing prices are around 1% above/below the trend each quarter. The multiplicative seasonal factors plot shows generally the same patterns as the additive, indicating that seasonal variation is consistent in level and proportion. Both plots indicate that housing prices follow a highly regular quarterly cycle, with predictable peaks and troughs that are constant over time, which supports the conclusion that housing prices in this series exhibit mild, stable seasonality rather than seasonality that scales with housing prices level.

f

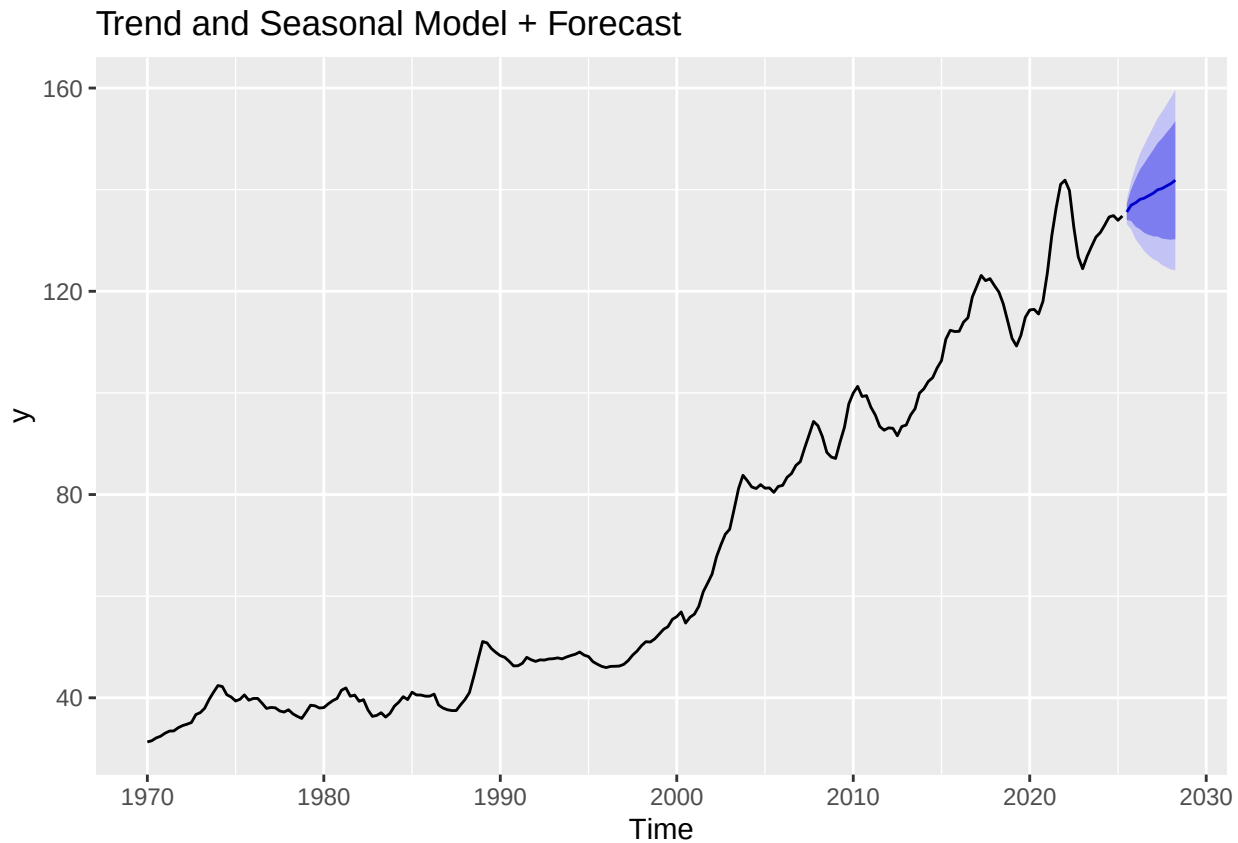
```
stl_fit <- stl(y, s.window = "periodic", robust = TRUE) ## Additive seasonal decomposition with STL
y_sa <- seasadj(stl_fit) ## Seasonally adjusted series

ar_fit <- auto.arima(y_sa, seasonal = FALSE, stepwise = FALSE, approximation = FALSE) #model

fc_sa <- forecast(ar_fit, h = 12) ##forecast 12 steps ahead

fc <- stlf(y, h = 12, s.window = "periodic", robust = TRUE, ## Re-seasonalize + add trend
           method = "arima")

autoplot(fc) + ggtitle("Trend and Seasonal Model + Forecast") ## Plot (data, fit, and forecast)
```



This model takes the additive seasonal decomposition with the trend/cycle components to forecast housing prices 12 quarters ahead. The plot shows the historical values and the 12-step forecasted values, with 80% and 95% confidence intervals marked by the shaded blue areas.

Conclusions and Future Work

The Final Model effectively captures the key characteristics of this quarterly housing price series. By decomposing the data additively into trend, seasonal, and cyclical/irregular components, the model isolates a smooth long-term growth trend and a stable, repeating quarterly seasonal pattern, which produces a 12-step forecast that extends the trend and seasonal effects. The forecasts show a continuation of the upward trend in housing prices, with quarterly seasonal fluctuations consistent with the historical data. Overall, the additive approach with fixed seasonality provides a strong fit for Real Residential Home Prices in Australia. In the future, this forecast could potentially be improved by utilizing a piece wise trend to more accurately model periods of slow or accelerated growth. Additionally, the addition of some external regressors like interest rates or inflation could improve the model's forecasting power, and finally, future analysis could pit this model against others to more formally address its accuracy.

References

All data was sourced from FRED financial data, specifically from this link: <https://fred.stlouisfed.org/series/QAUR628BIS>